Chapter 12 — Nuclear and Particle Physics

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Mass number A equals: (a) Z − N (b) Z + N (c) N − Z (d) Z only

2. Strongest force: (a) strong (b) gravity (c) electromagnetic (d) weak

3. Pair production minimum: (a) 931 MeV (b) 125 GeV (c) 0.51 MeV (d) 1.02 MeV

4. Alpha decay reduces A by: (a) 2 (b) 0 (c) 4 (d) 1

5. Beta-minus decay changes Z by: (a) 0 (b) −1 (c) +1 (d) +2

6. Proton quarks: (a) udd (b) uuu (c) uud (d) ddd

7. Baryons have: (a) 3 quarks (b) 4 quarks (c) q and $\bar{q}$ (d) 2 quarks

8. Higgs boson gives: (a) colour (b) mass (c) charge (d) spin

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is the strong force needed inside the nucleus?

2. Why do beta particles have a continuous energy spectrum?

3. Why are beta particles not orbital electrons?

4. Why is a nucleus needed for pair production?

5. Why are protons and neutrons not fundamental particles?

6. Why is the Higgs boson important in modern physics?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Compare alpha, beta and gamma radiations in nature, charge, ionization and penetration. **(4)**

(b) Explain beta-minus and beta-plus decay with examples. **(4)**

Q.4 (a) Classify fundamental particles into leptons, hadrons and exchange bosons. **(4)**

(b) Describe the quark model of protons and neutrons and beta decay at quark level. **(4)**

studysapiens.com